

Government Polytechnic, Dharashiv.

First Formative Assessment of Theory (FA-TH) Test AY-2024-25

Program Name: Computer Engineering Program (CO)

Semester : Third (CO3K)

Course Title : Data Structures Using C - 313301

Marks : 30

Duration: 1.5 Hour

Date : 28/08 /2024

CO N	Course Outcome (CO)
CO1	Perform Basic Operations on Array
CO2	Apply different searching and sorting methods.
CO3	Implement basic operations on Linked List.
CO4	Perform operations on Stack using Array and Linked list Implementation.
CO5	Perform operations on Queue using Array and Linked list Implementation.
CO6	Create and traverse Tree to solve Problems.

Instructions: Question 1 and 2 are compulsory.

Q1	Attempt any Five (2x5=10 Marks)	Marks	CO
a	Give classification of Data Structures?	2	1
b	Differentiate between Static and Dynamic Memory Allocation?	2	3
c	Explain terminologies Node, Address, Next Pointer, and Null Pointer?	2	3
d	List Operations on Data Structures?	2	1
e	Explain types of Linked list with diagram?	2	3
f	Define Abstract Data Type?	2	1
g	Explain with diagram circular linked list?	2	3

Q2	Attempt any Five (4x5=20 Marks)	Marks	CO
a	Write a C Program for Insertion and Display operations on Array?	4	1
b	Write Algorithm for Bubble Sort?	4	2
c	Write Algorithm for Binary Search?	4	2
d	Write a C program for creation of Linked list?	4	3
e	Write Algorithm for addition of new node in linked list at Beginning and End?	4	3
f	Write C program for linear search on Array?	4	1
g	Write an algorithm to count number of nodes in singly linked list?	4	3

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e	Write Algorithm for addition of new node in linked list at Beginning and End?	4	3
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